millionmessages.us — Setup Guide

Account-side steps to take the site from the files in this folder to a live, secured, backed-up service. October 20, 2026.

> The code is finished and tested locally. Everything below is configuration in accounts only you can log into. Budget about an hour, mostly waiting on DNS. Cloudflare's dashboard changes layout from time to time, so where a menu label differs from what's written here, look for the same concept.

0. What you are setting up

One Cloudflare Worker serves the website and runs the counter. There is no server to maintain. The pieces:

Piece	What it is	Where it lives
Website	The pages in `public/`	Cloudflare Worker (static assets)
Counter	One integer + 24-hour abuse-prevention hashes	Cloudflare Durable Object (free tier)
Human check	Cloudflare Turnstile widget on the "I sent it" button	Cloudflare dashboard
Secrets	`SALT_SECRET`, `ADMIN_SECRET`, `TURNSTILE_SECRET`	Cloudflare, encrypted; never in code
Backups	Nightly copy of the count	Your Mac mini
Source	Public repository	GitHub

1. Secure the Cloudflare account (10 minutes)

1. Log in at dash.cloudflare.com. Open your profile → **Authentication**.
2. Turn on two-factor authentication. Add **Security Key** authentication and register at least two: Touch ID on the Mac, plus a second key (a YubiKey, or Touch ID on a second Apple device). Cloudflare recommends more than one so you can't be locked out.

3. Download the **backup codes** and store them in your password manager.
4. If email-code authentication is enabled, turn it off once the keys work — it is the weakest factor.
5. Confirm `millionmessages.us` shows under **Domain Registration** and that auto-renew is on.

2. Install the command-line tool on your Mac (10 minutes)

Cloudflare deploys from a tool called Wrangler, which needs Node.js.

1. Install Node.js (LTS) from nodejs.org, or with Homebrew: `brew install node`.
2. Put the `millionmessages` folder somewhere permanent (for example `~/Projects/millionmessages`). Delete `node_modules` and `.wrangler` if they came along; they'll be rebuilt.
3. In Terminal:

   ```
   cd ~/Projects/millionmessages
   npm install
   npx wrangler login
   ```

   The last command opens a browser window to authorize Wrangler against your Cloudflare account.

3. Create the Turnstile human check (5 minutes)

1. Dashboard → **Turnstile** → **Add widget**.
2. Name: `millionmessages`. Hostnames: `millionmessages.us` and `www.millionmessages.us`. Widget mode: **Managed**.
3. Copy the two keys it gives you: the **Site key** (public) and the **Secret key** (private).
4. Paste the **site key** into `public/config.js` as the value of `turnstileSiteKey`.

> The embed on other websites loads inside an iframe from millionmessages.us, so the Turnstile hostname list above covers it — you never need to add other people's domains.

4. Set the secrets (5 minutes)

Generate two long random strings in Terminal (run it twice):

```
openssl rand -base64 32
```

Then store all three secrets in Cloudflare. Each command prompts you to paste the value; it is encrypted and never appears in the code or the repository.

```
npx wrangler secret put SALT_SECRET        # first random string
npx wrangler secret put ADMIN_SECRET       # second random string — keep a copy
in your password manager
npx wrangler secret put TURNSTILE_SECRET   # the Turnstile SECRET key from step 3
```

> If the Worker has not been deployed yet, Wrangler may ask whether to create it when you set the first secret. Answer yes. Keep `ADMIN_SECRET` in your password manager: it's the only way to restore the count from a backup.

5. Deploy (5 minutes)

```
npx wrangler deploy
```

Because `wrangler.toml` lists `millionmessages.us` and `www.millionmessages.us` as custom domains, Cloudflare creates the DNS records and certificate automatically. The site is usually reachable within a minute or two; occasionally DNS takes longer.

Check:

- `https://millionmessages.us/api/health` → `{"ok":true}`
- `https://millionmessages.us/api/count` → `{"count":0,"updated":null}`
- `https://millionmessages.us/` → the page, with the Turnstile checkbox visible above "I sent it".

6. Test the real thing (10 minutes)

1. On the live site: click **Copy the message**, open the White House form, actually send it, come back, pass the human check, click **I sent it**. Count becomes 1.
2. Reload: the button should read "Already counted today". Click again from your phone on cellular (a different IP): count becomes 2.

3. Try the embed: make a plain text file `test.html` containing only `<script src="https://millionmessages.us/embed.js" async></script>` and open it in a browser. The compact widget should appear with the live count.
4. Reset to zero before launch (the two test clicks shouldn't count):

```
ADMIN_SECRET='paste-it-here' scripts/restore.sh 0
```

7. Backups on the Mac mini (10 minutes)

The count is the only state the site has, so backing it up means saving one number once a day.

1. Edit `scripts/us.millionmessages.backup.plist` and replace every `REPLACE_WITH_FULL_PATH` with the real path (for example `/Users/will/Projects`).
2. In Terminal, from the project folder:

```
scripts/backup.sh                                    # run once by hand;
should print "backed up count=…"
cp scripts/us.millionmessages.backup.plist ~/Library/LaunchAgents/
launchctl load ~/Library/LaunchAgents/us.millionmessages.backup.plist
```

3. It now runs every night at 3:17 am and writes to `backups/`. Check `backups/history.csv` after a few days.

To restore after any problem: `ADMIN_SECRET='…' scripts/restore.sh` (uses the latest backup) or `scripts/restore.sh 123456` for a specific number.

8. Publish the source on GitHub (15 minutes)

1. Create a free GitHub account under your own name if you don't have one; enable two-factor there too.
2. Create a new **public** repository named `millionmessages`. Don't add a README (one exists).
3. From the project folder:

```
git init
git add .
git commit -m "Initial public release"
git branch -M main
```

```
git remote add origin https://github.com/YOUR-USERNAME/millionmessages.git
git push -u origin main
```

`.gitignore` already excludes `.dev.vars`, `node_modules`, `.wrangler` and `backups`, so no secret or backup is ever pushed. Check the repository page to be sure.

4. Put the repository URL into `public/config.js` as `sourceUrl`, redeploy with `npx wrangler deploy`, and the "Source code on GitHub" link on the site goes live.

> Optional, stronger transparency: in the Cloudflare dashboard, open the Worker → Settings → Build and connect it to the GitHub repository. Each deployment is then tied to a public commit, so anyone can see that what's running is what's published.

9. Before you tell anyone (checklist)

- ☐ Count reset to 0 (step 6.4) and the two test clicks are gone.
- ☐ The message in `config.js` is your final wording, and `versionHistory` has its first entry dated today.
- ☐ `sourceUrl` set; the GitHub link on the page works.
- ☐ `hello@millionmessages.us` exists — set up **Email Routing** in Cloudflare (free) to forward it to an address you read, or change the footer address.
- ☐ Backup ran at least once by hand and once overnight.
- ☐ Security keys registered on Cloudflare and GitHub; backup codes in the password manager.

Changing the message later

Edit `public/config.js` (message text and a new `versionHistory` line), then `npx wrangler deploy`. The counter is untouched; every embed on other sites updates automatically.

If something goes wrong

Symptom	Likely cause / fix
"I sent it" says server-not-configured	`SALT_SECRET` not set — step 4.
Turnstile widget never appears	Site key missing from `config.js`, or hostname not listed in the Turnstile widget.

Click returns "turnstile-failed"	`TURNSTILE_SECRET` wrong or belongs to a different widget.
Count looks wrong or was tampered with	Restore from backup (step 7); then rotate `ADMIN_SECRET` with `wrangler secret put`.
Site unreachable right after deploy	DNS/certificate still propagating; wait 10–15 minutes.